Latihan Soal Integral Tentu

No.

Date:

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Bagian A

Hitunglah integral tentu :

a). $\int_{-1}^{2} (3x^2 - 2x + 3)\, dx$

Hitung dulu integral tak tentu

$\int 3x^2 - 2x + 3\, dx$

$\int f(x) \pm g(x)\, dx = \int f(x)\, dx \pm \int g(x)\, dx$

$\int 3x^2\, dx - \int 2x\, dx + \int 3\, dx$

$= x^3 - \int 2x\, dx + \int 3\, dx$

$= x^3 - x^2 + \int 3\, dx$

$= x^3 - x^2 + 3x$

$= (x^3 - x^2 + 3x)\Big|_{-1}^{2}$ → gunakan $F(x)\Big|_a^b = F(b) - F(a)$

$= 2^3 - 2^2 + 3 \times 2 - \left((-1)^3 - (-1)^2 + 3 \times (-1)\right)$

$= 8 - 4 + 3 \times 2 - (-1 - 1 + 3 \times (-1))$

$= 15$

b. $\int_1^2 (x+1)^2\, dx = \int_1^2 x^2 + 2x + 1\, dx$

$= x^3/3 + x^2 + x + c\,\Big|_1^2$

$= \frac{8}{3} + 4 + 2 + c - \left(\frac{1}{3} + 1 + 1 + c\right)$

$= \frac{8}{3} + 6 + c - \left(\frac{1}{3} + 2 + c\right)$

$= \frac{26}{3} + c - \frac{7}{3} + c = \frac{19}{3}$

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8. Bagian B

Gambarkan grafik fungsi berikut, dan hitung harga rata-rata dan efektif f dalam selang yang diberikan :

1. $F(x) = 9 - x^2 \;;\; (-3, 3)$

$$y_{av} = \frac{1}{3-(-3)} \int_{-3}^{3} (9 - x^2)$$

$$= 1/6 \left| \begin{matrix} 3 \\ -3 \end{matrix} \right. \frac{-x^3}{3} + 9x + C$$

$$= \frac{-x^3}{18} + \frac{9}{6}x + \frac{C}{6} \left| \begin{matrix} 3 \\ -3 \end{matrix} \right.$$

$$= \frac{-(3)^3}{18} + \frac{9}{6}(3) + \frac{C}{6} - \left(\frac{-(-3)^3}{18} + \frac{9}{6}(-3) + \frac{C}{6} \right)$$

$$= \frac{-27}{18} + \frac{27}{6} - \frac{27}{18} + \frac{27}{6} = 2\left(\frac{3}{2}\right) + \frac{54}{6} = -3 + 9 = 6$$

$$y_{ev} = \sqrt{\frac{1}{6} \int_{-3}^{3} (9 - x^2)^2 \, dx} = \sqrt{1/6 \int_{-3}^{3} (x^4 - 18x^2 + 81 \, dx}$$

$$= \sqrt{1/6 \left| \begin{matrix} 3 \\ -3 \end{matrix} \right. (x^5/5 - 6x^3 + 81x + C}$$

$$= \sqrt{1/6 \left| 3^5/5 - 6(3)^3 + 81(3) + C \left(\frac{3}{5}\right)^5 \right.}$$

$$-6(-3)^3 + 81(-3) + C$$

$$= \sqrt{1/6 \times 2 \left(3^5/5\right) - (2)\,6(3)^3 + 2(81)(3)}$$

$$= \sqrt{(1/6)\left(\frac{1296}{5}\right)} = \sqrt{\frac{216}{5}} = \frac{6\sqrt{30}}{5}$$

2. $F(x) = x, \; 0 \le x < \pi$

$$y_{av} = \frac{1}{\pi} \int_{0}^{\pi} x \, dx = \frac{1}{\pi} \left(\frac{x^2}{2} \right) + C \left| \begin{matrix} \pi \\ 0 \end{matrix} \right.$$

$$= \frac{1}{\pi} \left(\frac{\pi^2}{2} + C \right) - \left(\frac{0}{2} + C \right)$$

$$= \frac{1}{\pi} \left(\frac{\pi^2}{2} \right) - \frac{\pi}{2}$$

$$y_{ev} = \sqrt{\frac{1}{\pi} \int_{0}^{\pi} x^2 \, dx} = \sqrt{\frac{1}{\pi} \int_{0}^{\pi} \frac{x^3}{3} \, dx}$$

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$$= \sqrt{\frac{1}{\pi}\frac{\pi^3}{3}} = \sqrt{\frac{\pi^2}{3}} = \frac{\pi}{\sqrt{3}}$$

Bagian C

Gambarkan grafik fungsi periode 2π berikut

5. $f(x) = \begin{cases} (x+2) & -\pi < x < 0 \\ (-x+2) & 0 < x < \pi \end{cases}$

$$y_{ev} = \frac{1}{2\pi}\left(\int_{-\pi}^{0} (x+2)^2\,dx + \int_{0}^{\pi} (-x+2)^2\,dx\right)$$

$$= \frac{1}{2\pi}\left(\int_{-\pi}^{0} x^2 + 4x + 4 + (x^2 - 4x + 4)\,\Big|_0^{\pi}\right)$$

$$= \frac{1}{2\pi}\left(-(\pi^2 + (-4\pi) + 4) + (\pi^2 - 4\pi + 4)\right)$$

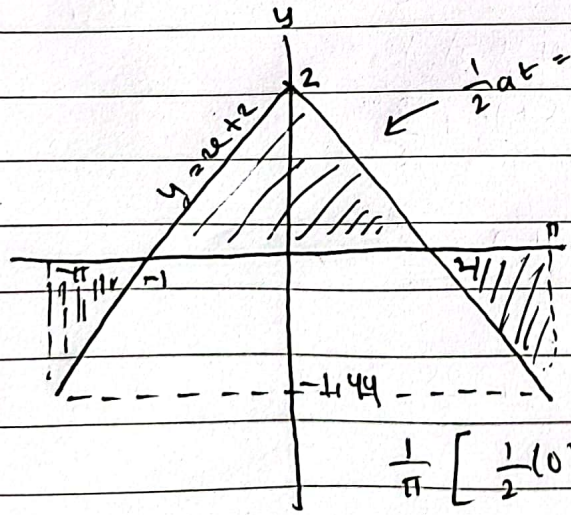
$$= \frac{1}{2\pi}\left(-\pi^2 + 4\pi + 4 + \pi^2 - 4\pi + 4\right)$$

$$= \frac{1}{2\pi}(0) = 0$$

$$y_{ev} = \frac{1}{2\pi}\left(\int_{-\pi}^{0} (x^2+2)\,dx + \int_{0}^{\pi} (-x+2)\,dx\right)$$

$$= \frac{1}{2\pi}\left(\frac{x^2}{2} + 2x\,\Big|_{-\pi}^{0} + \left(\frac{-x^2}{2} + 2x\right)\Big|_0^{\pi}\right)$$

$$= \frac{1}{2\pi}(-\pi^2 + 4\pi) = \frac{-\pi}{2} + 2$$



$\frac{1}{2}at = \frac{1}{2}\cdot 2\cdot 2 = 2$

$\frac{1}{2}a.t = \frac{1}{2}$

$$\frac{1}{\pi}\left[\frac{1}{2}(0)^2 + 2(0) - \left(\frac{1}{2}(-\pi^2) + 2(-\pi)\right)\right]$$

$$= \frac{1}{\pi}(0 - (-11).21)) = 3,57$$

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6. $F(t) = -t^2 \quad -\pi < t < 0$

$t^2 \quad 0 < t + \pi$

$$\frac{1}{2\pi}\left(\int_{-\pi}^{0} -t^2\,dt + \int_{0}^{\pi} t^2\,dt\right)$$

$$1/2\pi\left(-\left(-\frac{\pi^3}{3}\right) + \left(\frac{\pi^3}{3}\right)\right)$$

$$\frac{1}{2\pi}\left(\frac{2\pi^3}{3}\right) = \frac{2\pi^3}{3}\left(\frac{1}{2\pi}\right) = \frac{\pi^2}{3} = y_{ev}$$

$f(t) = \sqrt{-t^2 \quad -\pi < t < 0}$

$\sqrt{t^2 \quad 0 < t < \pi}$

$$\sqrt{\frac{1}{2\pi}\left(\int_{-\pi}^{0} -t^4\,dt + \int_{0}^{\pi} t^4\,dt\right)}$$

$$\sqrt{\frac{1}{2\pi}\left(-\frac{(-\pi)^5}{5} + \left(\frac{\pi^5}{5}\right)\right.}$$

$$\sqrt{\frac{1}{2\pi}\left(2\left(\frac{\pi^5}{5}\right)\right)}\sqrt{=\frac{\pi^4}{5}} = \frac{\pi^2}{\sqrt{5}}$$

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